

Sandeep Sagar Madanu

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ABOUT

Data Scientist & AI/ML Engineer specializing in **predictive modeling**, **statistical inference**, **deep learning**, and **LLM-powered applications**. Proficient in end-to-end **analytics pipelines**, **ML lifecycle management**, **interactive visualization**, and **prompt-driven AI products**. Currently directing independent research in **explainable AI (XAI)** and **Vision Transformers** for radiology. Open to **Data Science**, **AI/ML Engineering**, **Data Analytics**, and **Prompt Engineering** roles.

TECHNICAL SKILLS

Programming & Querying:	Python, SQL, Jupyter, Git/GitHub
ML / Deep Learning:	Scikit-learn, TensorFlow, Keras, PyTorch, CNN, ViT, Feature Engineering, Hyperparameter Tuning, Cross-Validation
Data Science & Analytics:	Pandas, NumPy, EDA, Statistical Analysis, Predictive Modeling, A/B Testing Concepts, KPI Reporting
NLP, LLMs & Prompt Engineering:	LangChain, LangGraph, Hugging Face, LLaMA, GROQ API, RAG, Agentic AI, Prompt Design, Tool-Use Orchestration
Computer Vision & XAI:	OpenCV, MediaPipe, Grad-CAM, SHAP, Radiomics, Image Classification
Visualization & Deployment:	Matplotlib, Seaborn, Streamlit, FastAPI, REST APIs, Model Deployment
Domains:	Medical Imaging AI, Autonomous Systems, Conversational AI, Predictive Analytics

EXPERIENCE

AI Researcher — Vision Transformers & Medical Imaging

Dec 2025 – Present

Self-Employed (Independent Research)

Hyderabad, India

- Directed investigation of **hybrid CNN–Vision Transformer** architectures for **MRI-based MGMT prediction** on 500+ scans, pursuing 1 Scopus-indexed publication in Medical AI / XAI.
- Constructed **multi-modal ingestion frameworks** unifying 3 input streams (MRI, radiomics, clinical records) via **5-fold CV**, feature extraction, and reproducible benchmark suites.
- Benchmarked explainability with **Grad-CAM** and **SHAP** over 5 hold-out folds, assessing attention heatmap alignment with tumor boundaries for clinical reliability.

Data Science Intern

Apr 2025 – Jun 2025

Brainovision Solutions India Pvt. Ltd.

Hyderabad, India

- Produced 3+ Python **ML workflows** spanning ETL, model fitting, and performance audits on production datasets surpassing 10K rows across diverse industry sectors.
- Collaborated on 2 **forecasting analytics** engagements in 2-week Agile cycles, generating exploratory analyses and leaderboard comparisons that lifted baseline macro-F1 by 8–12%.

Project Intern — AI & Autonomous Systems

May 2024 – Sep 2024

Aigen Labs (Aeroforge Labs)

Hyderabad, India

- Parsed sensor and telemetry streams from 2 UAV and **autonomous-vehicle** prototypes to fuel automation logic and vision-based proof-of-concepts.
- Built low-latency **vision subsystems** for robotic pick-and-place operations, sustaining 25+ FPS throughput on embedded unmanned platforms.

PROJECTS

Multimodal Image Query System — LLM & Prompt Engineering

Python, GROQ API, LLaMA, FastAPI

- Launched a **vision-language retrieval platform** leveraging **LLaMA** via **GROQ API**, exposing 10+ **FastAPI** REST routes with iteratively refined prompt templates.

Agentic AI Chatbot with LangGraph

Python, LangGraph, LangChain

- Created a **conversational agent** in LangGraph and LangChain, executing 5+ chained tool invocations per dialogue turn through persistent state graphs.

Medical Image Classification (Brain Tumor & Pneumonia Detection)

Python, TensorFlow/Keras, CNN

- Fine-tuned 2 **CNN classifiers** on 3,000+ brain MRI and thoracic X-ray samples, reaching 92%+ hold-out accuracy and 0.89 macro-F1 for triage assistance.

Driver Drowsiness Detection & Traffic Sign Recognition

Python, OpenCV, MediaPipe, CNN, Scikit-learn

- Shipped a driver-alertness tracker at 30 FPS via MediaPipe facial landmarks and a **43-class roadway sign detector** at 96% test accuracy for ADAS applications.

EDUCATION

PGDM, Artificial Intelligence & Data Science

2023 – 2025

ASB — Ashoka School of Business

Hyderabad, India

B.Com in Commerce & Computer Applications

2020 – 2023

Andhra Loyola College

Vijayawada, India

CERTIFICATIONS

Python Using AI Workshop | Python Machine Learning 5-Day Bootcamp | AI and Data Scientist Certification